

A Comparative study on Electric Vehicle and Internal Combustion Engine Vehicles

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Abstract—*Electrification of the vehicles gains a significant research importance due to the increasing amount of greenhouse gas (GHG) emission by using conventional internal combustion engine vehicles (ICEV). The major benefits of electric vehicles (EVs) are a reduced amount of carbon monoxide (CO) emissions, higher efficiency, performance, and lower maintenance costs. It also allows energy diversification by switching to renewable resources. This paper analyzes the efficiency of an EV with ICEV by considering various parameters like required torque, speed and distance traveled. The simulation is developed in MATLAB/Simulink and analysis is presented for various performance metrics as a measure.*

Keywords – *Powertrain modeling, electric vehicle, internal combustion engine vehicle, electric motor, greenhouse gases, constant tractive effort, variable tractive effort.*

I. INTRODUCTION

Rapid urbanization, increased greenhouse gas emission, and over usage of existing resources have raised the alarming need for better and sustainable solutions for the transportation sector [1]. The conventional internal combustion engine (ICE) vehicles are proven to be less energy efficient and more in GHG emission. Electrification of vehicles can improve these issues and also helps in reducing the usage of fossil fuels [2].

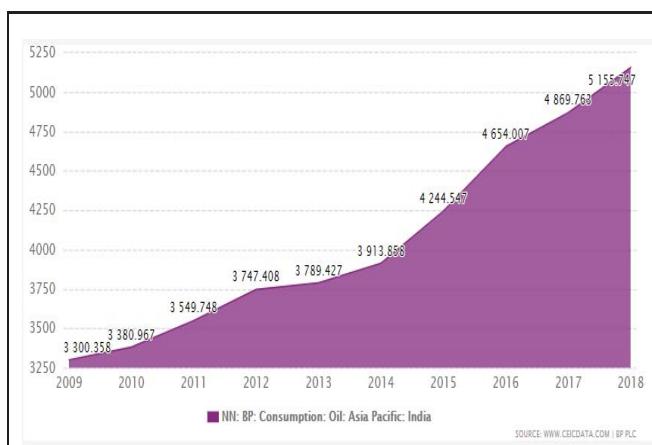


Fig. 1. Oil consumption in India [6]

The oil consumption rate in India is shown in Fig. 1. In the case of ICEVs the cost spent on fuel consumption during the overall ownership of the vehicle comes up to a massive sum and as per the statistics collected by Census and Economic information Centre (CEIC), the oil consumption of the country has reached an amusing number of 5,155,747 barrels, most of which are imported from other countries. The costs spent on fossil fuels have

been increasing rapidly in the past few years [3]. The increased electricity costs due to EV usage can be managed efficiently using Smart Grid Technology (SGT) which reduces the energy wastage by communication and keeps it to a minimum even during peak hours [4].

The fuel consumption of the vehicle is an important parameter to understand as it depends on various real-time parameters like vehicle type, road conditions, aerodynamic drag, and power train along with system-level specifications of the vehicle [5].

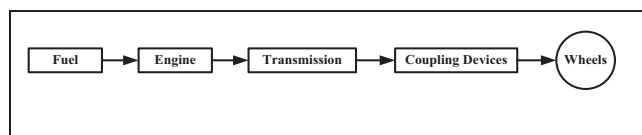


Fig. 2. The Block diagram depicts an internal combustion engine vehicle (ICEV)

The block diagram of the ICE vehicle and Electric vehicle is presented in Fig. 2 and Fig. 3 respectively. The combustion of fuel generates the required power and is transmitted to the drive system via transmission and coupling devices. Losses due to transmission take the major part in an ICEV. The fuel conversion efficiency of an ICE vehicle is limited to 25% only [6].

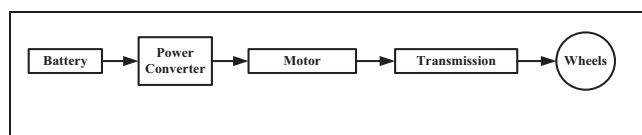


Fig. 3. The Block diagram depicts the electric vehicle system

In Fig. 3, the battery supplies required regulated power for the motor through the power converter. Power converters used in EV has higher efficiency. Number of mechanical parts involved in the power conversion process from input to wheels of electric vehicles are less which intend eliminates huge mechanical losses when compared to ICEV [7].

Fig. 4, shows the requirement of complex gear for the IC Engine. The complex gear system in ICEV is due to its narrow speed range operation [8]. A wide range of gear ratios are used to meet the torque requirement by the vehicle at different driving conditions. An ICEV cannot shoot up to 6000 rpm from 0 rpm without using any gears due to its design limitations. Due to multiple gear ratios, the problem of mechanical loss comes into the picture and reduces the efficiency of the system. But in EVs, a simple transmission gear is required as the motor can rise from 0 rpm to rated rpm. But to maintain low speeds an extra gear

may be required. When compared with its IC counterparts, EV has a lesser number of gears which is another factor that contributes to the higher efficiency of the vehicle [9].

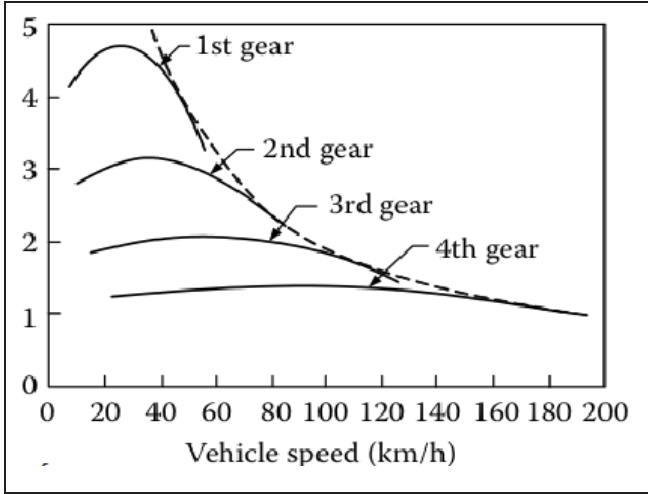


Fig. 4. Complexity in the gear of an internal combustion engine vehicle (ICEV) [3]

II. TRANSMISSION SYSTEMS

A. Transmission of an EV

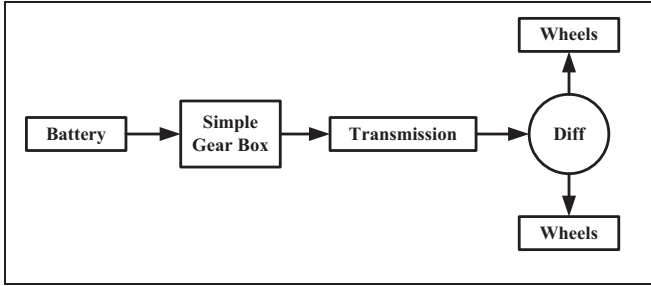


Fig. 5. The powertrain of an Electric Vehicle (EV)

The Power generated at the motor is transferred through a simple gearbox to the transmission and the wheels of EV is shown in Fig5. It doesn't require a complex construction because of its variable torque-speed characteristics. There exist different configurations of EVs based on the position of the motors, battery/energy source, number of motors used. The selection of the configuration is done based on the type of vehicle and application. The battery pack chosen must satisfy the power requirements of motor. This also has to be developed and modeled to estimate different parameters of battery [10].

The complexity of the system is widely reduced in EVs due to the elimination of various mechanical parts such as crankshaft, gear wheels, clutch, and differential axles. The usage of digital systems improves the accuracy of the vehicle in terms of power transmission.

The band of speed range is significantly higher for an EV and this allows us to reach higher speeds in a few seconds. Though this highly depends on the power of the motor used and the capacity of the battery. The initial cost of an EV is higher than an ICEV. But this comes with reduced maintenance costs. The main drawback of EV being underdeveloped are infrastructure for charge stations and battery units. This can be solved using methods like move and charge technique and on line power transfer [11].

B. Transmission of an ICE Vehicle

The power generated is transferred to a torque converter and thereby to a complex gearbox which is separated by a clutch for the gear shifting [12]. Transmission of power from the engine to wheels involves many intermediary mechanical parts thus reducing the efficiency of the vehicle [13].

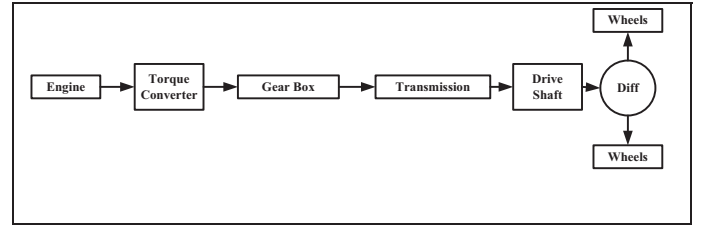


Fig. 6. The powertrain of an internal combustion engine vehicle (ICEV)

III. RESULT AND ANALYSIS

A. Vehicle dynamics

Characteristics of the vehicle have been studied to define the performance of the vehicle under various circumstances. Vehicle's output torque, power, and speed are a few of the most important characteristics which define the efficiency and performance of the vehicle. Thus, comparing these characteristics would provide us a brief idea of EV and ICEV work under similar conditions and drop us down to a conclusion based on the comparison done with the obtained results using MATLAB/Simulink. The vehicle model considered has a mass of 1500kg, a maximum torque of 250Nm, and a power of 100kW.

B. Constant Tractive effort

In the case of constant tractive effort methods, not all practical conditions are taken into consideration while plotting the graphs. This gives an idea about how the system inside the vehicle acts with varying speeds. The motor output torque and its power are considered to be constantly efficient and speeds are varied w.r.t time to obtain the results.

The motor manages to drive the vehicle at constant torque and speed modes and maximum constant power is obtained only after constant speed mode. As shown in Fig. 7, the power reaches maximum level in 50s, for a profile where the tractive effort is constant. Power consumption can be directly related to the size and specifications of the motor required and in turn to the capacity of the battery needed to run the motor which helps us in designing the charger required to charge the battery. The forces contributing to the tractive effort in a vehicle is the aerodynamic drag (F_{ad}), rolling resistive force (F_{rr}), gradient force (F_{gd}), and acceleration force (F_{aa} and F_{la})

$$F_{te} = F_{ad} + F_{rr} + F_{gd} + F_{aa} + F_{la} \quad (1)$$

$$F_{ad} = 0.5 * A * \rho * C_d * (v_{air} + v) \quad (2)$$

$$F_{rr} = \mu_r * m * g * \cos(\Theta) \quad (3)$$

$$F_{gd} = m * g * \sin(\Theta) \quad (4)$$

$$F_{la} = m * a \quad (5)$$

$$F_{aa} = \left(\frac{J}{r_2}\right) * \frac{dv}{dt} \quad (6)$$

Equation (2) depends on the air density (ρ), coefficient of drag (C_d), velocity of the air (v_{air}), and velocity of the vehicle (v). (3) Depends on the coefficient of friction (μ_r), mass of the vehicle (m), and the gradient (Θ). Equation (4)

depends on the mass of the vehicle and the gradient. (5) Depends on the mass and acceleration (a) of the vehicle whereas (6) of the vehicle depends on the moment of inertia (J) of the vehicle, radius of the wheel (r), and the acceleration of the vehicle. Fig. 8, shows the distance covered for a constant tractive effort profile, and it covers 10m in 150sec.

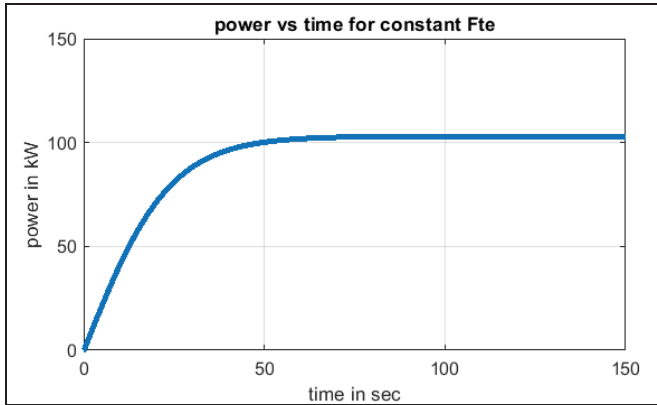


Fig. 7. Power vs. time for constant tractive effort

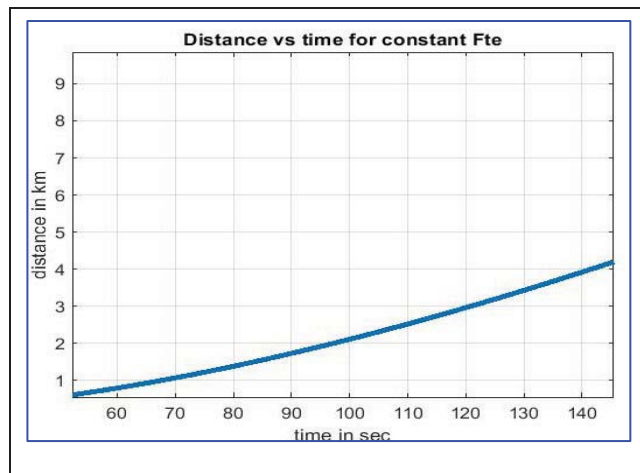


Fig. 8. Distance vs. time for constant effort

The speed at the starting of the vehicle is low. The maximum speed considered for the model is 150kmph, and this is reached in 30sec as depicted by Fig. 9. But here the torque developed is high and is equal to the rated torque of the motor to overcome the transients.

The main advantage of an EV is that it provides full torque in no time and thus helps in overcoming opposing forces faster than in ICEVs.

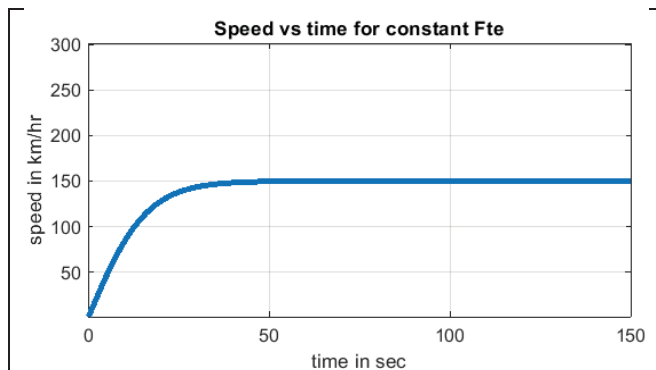


Fig. 9. Speed vs. time for constant effort

C. Variable Tractive Effort

This takes into consideration, not only the constant tractive effort but also the starting tractive effort, the maximum tractive effort, and the continuous tractive effort. Here the maximum effort is put in the starting where it takes 0.5 sec for a vehicle of mass 1500kg, with a maximum motor torque of 250Nm and power of 100kW to overcome the initial transients acting on the vehicle and initiate the motion of the movement further. This mode of operation is the constant torque mode of operation after which comes the constant speed mode operation until it overcomes its rated speed. This model takes 2.4sec to overcome the rated speed. Upon further acceleration, constant power mode comes into the picture where no further acceleration is provided and the speed remains to be constant maximum speed and thus maintaining the torque constant.

$$T_w = F_{te} * r \quad \text{and} \quad w_w = \frac{v}{r}, G = \frac{w_m}{w_w} \quad (7)$$

$$T_m = \left(\frac{1}{G} * n_g\right) * T_w \quad (8)$$

The torque of the wheel (T_w) is found using the total tractive effort (F_{te}) and the radius of the wheel. Gear ratio (G) and efficiency of gearing (n_g) play an important role in determining the motor torque (T_m).

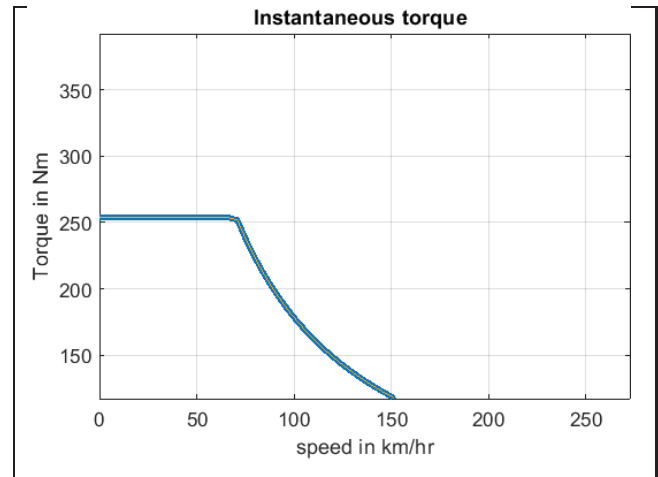


Fig. 10. Torque vs. time for variable effort

In the above Fig. 10, the motor takes no time to reach the maximum torque when compared with an ICEV. By observing the graphs of both ICEV and EV for Torque vs. Speed and Torque vs. RPM in case of ICEV of variable tractive effort, it can be noticed that the vehicle gains maximum torque at starting. The maximum speed of the vehicle is considered to be 150 kmph and the rated speed of the vehicle is considered to be 70 kmph. When the vehicle overcomes the transients and inertia, it reaches the maximum speed mode where there is no acceleration and a constant maximum speed of 150 kmph is maintained until brakes are applied.

The above model in Fig. 11, is developed in MATLAB/Simulink by taking the dynamo-meter reference application [11]. The Engine system represents the components of ICEV such as the engine and gearbox. The engine system is re-calibrated to our required system specifications and engine power is chosen as 100kW which is the same in the case of EV.

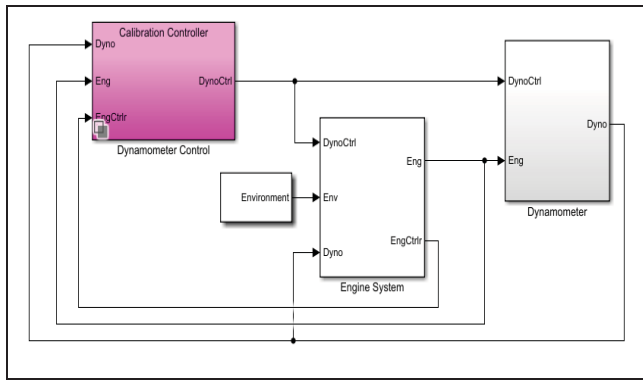


Fig. 11. Engine Dynamo-meter model [10]

RPM range is set to (0-5000). A dynamometer is a device used in measuring torque and rotational speed (RPM) of an engine simultaneously [14]. Steady-state results are plotted.

The dynamometer results of the SI engine for the model shown in Fig.11 is presented in Fig.12. The difference in the colors represents the torque command of the vehicle. The torque produced at initial stages is raised gradually w.r.t the engine speed and reaches maximum torque condition at around 1964 rpm. This happens as the fuel must be delivered to the engine to produce power and many intermediate power transfers happen simultaneously between the gearbox, clutch, and differential which delay the power delivery. Dropping and rising of the torque takes time in ICEV which reduces the efficiency of the engine as a whole but whereas in EV the max torque is available at the initial stage (0 rpm) thus increasing the initial performance of the vehicle.

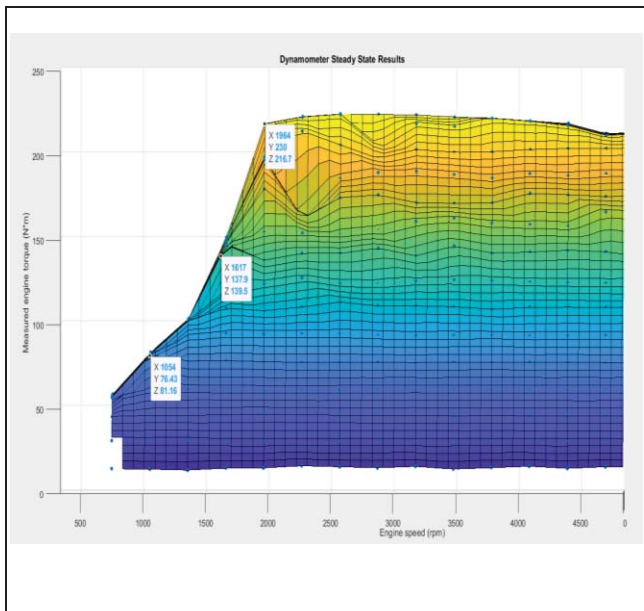


Fig. 12. Engine torque vs speed (RPM) [10]

Table. 1. COMPARISON BASED ON ANALYSIS [4][6]

Specification	ICEV	EV
Maximum torque	1964 RPM	0 RPM (instantaneous)

Emissions	Greenhouse gases	Zero
Efficiency	Low	High
Gearbox	multi-speed	single speed
Moving parts (in an engine)	2000 (approx.)	20 (approx.)
Maximum speed reach	multiple gears	single gear
Maximum RPM	13,000 - 18,000	6,000 - 8,000

IV. CONCLUSION

The comparison of ICEV with EV is done taking various important aspects such as torque, emissions, power, and Engine efficiency into consideration. In a comparison of characteristics with similar features, maximum torque in ICEV is reached at 1964 rpm, whereas in EV it is readily available at 0 rpm. So EV has better benefits of high starting torque to overcome the various forces acting on the vehicle. The maximum speed for an EV is achieved at 2.4 seconds which is due to its ability to overcome the transients quickly with the readily available initial high torque.

The power conversion efficiency of ICEV is observed to be less as the power from the engine has to be transferred through various channels before reaching the wheels, due to these moving parts mechanical losses are huge. But in case of EV, power is transferred to wheels through a simple channel which reduces the mechanical losses thus increasing the efficiency of transmission emissions of greenhouse gasses are negligible as the electric vehicle system is powered using a battery source which can be charged using renewable energy sources like solar energy. However, building an infrastructure for charging the vehicles is a huge task. Various other methods like wireless power transfer and battery swapping can be preferred to the conventional way of charging, yet it is expensive [15].

REFERENCES

- [1] Juyal, S.Sanjeevi, H.Saxena, A.Sharma, S.Singh, A.Chander, &Jhunjhunwala.A. , " Zero Emission Vehicles(ZEVs): Towards a policy fraework", NITI Aayog Report, pp.1–20, 2018.
- [2] Dr. Thirumalini S. and Ramasubramanian.A. "Investigation on use of Plug-in Hybrid Electric Vehicle (PHEV) Technology Using Renewable Energy for an Autorickshaw", Journal of KONES Powertrain and Transport, vol. 19, pp. 383 – 394, 2012.
- [3] ThomasD.Gillespie, "Fundamentals Of vehicle dynamics", Society of automotive engg, pp. 79-123, 1998.
- [4] "Electric Vehicle Transmission System – EV Transmission, Gears, Powertrain", India Renewable Energy Consulting - Solar, Biomass, Wind, Cleantech, 2019.
- [5] L. Grackova and I. Oleinikova, "Economic motivation for electric vehicles participation in power market, " 11th International Conference on the European Energy Market (EEM14), Krakow, pp. 1-5, 2014.
- [6] Julian Happian-Smith, "An introduction to Modern Vehicle Design", Butterworth-Heinemann, pp.157-184, 2002.
- [7] J. Shin et al, "Contactless power transfer systems for On-Line Electric Vehicle (OLEV), " IEEE International Electric Vehicle Conference, Greenville, SC, pp. 1-4, 2012.
- [8] A. A. Abulifa, R. K. R. Ahmad, A. C. Soh, M. A. M. Radzi and M. K. Hassan, "Modelling and simulation of battery electric vehicle by using MATLAB-Simulink, " IEEE 15th Student Conference on Research and Development (SCORED), Putrajaya, pp. 383-387, 2017.

- [9] P. Vishnu, Ajaykrishna, R., and Dr. Thiru.malini S., “Cost effective energy solution with dual battery for plug-in hybrid electric vehicle (PHEV) in a public transportation system”, 7th IEEE Conference on Industrial Electronics and Applications, ICIEA 2012, Singapore, pp.155-156, 2012.
- [10] D. Thiruvonasundari and K. Deepa, “Electric vehicle battery modelling methods based on state of charge– review,” J. Green Eng., vol. 10, no. 1, pp. 24–61, 2020.
- [11] “SI engine Dynamometer Reference Application”, Mathworks by Matlab
- [12] B.vijay and shanmugasundaram, R., “control strategy for parallel hybrid electric vehicles”, inprocedia computer science, vol.143, pp.678-685, 2018.
- [13] A.ramasubramanian, Dr.ThirumaliniS, Padmanaban, V.n.srinivasan, M., “Safe, Sustainable and smart charging of plug in Hybrid Electric Vehicles (PHEVs) ”, World Academy of Science, Engineering and Technology, vol.69, pp.1050-1055, 2012.
- [14] Tom Denton, “Automobile Mechanical and Electrical Systems, Automotive Technology: Vehicle maintenance and Repair”, Elsevier Ltd, pp.259-287, 2011.
- [15] V. Raveendran, Krishnan, N. N., and Dr. Manjula G. Nair, “Vehicle-to-grid Support by Electric Vehicle Charging Stations Operated at Airports and Metro Rail Stations”, in 2017 International Conference on Technological Advancements in Power and Energy (TAP Energy), 2017.